

DIGITAL DICE



Srajan Gupta (24BEC0066) | Devansh Pandey(24BEC0312)
| Surya Pratap Singh(24BML0069)

Prof.Dr. | Dhanabal R SENSE

Technology stack

Motivation/ Introduction

The **Digital Dice** is an innovative electronic circuit designed to replicate the random number generation of a physical dice using digital logic principles. It employs digital components such as flip-flops and counters to generate random numbers between **1 and 6**, providing a modern and efficient alternative to mechanical dice. The circuit is activated through a **push-button**, which triggers the random number generation process, and the result is displayed using an **LED-based pattern** that visually represents the face of a traditional dice. Its **purely digital design** ensures low power consumption, compact size, and reliable performance.

Hardware Description Language: Verilog HDL

Simulation Tool: ModelSim , Quartus Prime

Hardware Platform: Breadboard

Design Logic: Logic Gates

Display Interface: Seven-Segment Display Cathode

Clock Control: Clock Divider & Counter Logic

SCOPE of the Project

The scope of the **Digital Dice** project extends to the design, implementation, and demonstration of a fully digital system capable of generating random numbers between **1 and 6**, similar to a traditional dice. The project focuses on utilizing **digital logic components** such as flip-flops, counters, and display drivers to achieve random number generation and LED-based display output.

This project can be further expanded to include **microcontroller-based control**, **seven-segment display interfaces**, or **programmable randomization algorithms** for enhanced accuracy and versatility. It also provides a foundation for understanding **digital system design**, **clock pulse generation**, and **probability-based applications** in electronics.

Results

The system performed efficiently with a response time of less than 50 milliseconds from button press to display, a power consumption of approximately 180 mW (within the specified limits), and a 100% success rate across more than 500 test rolls. The randomness quality was excellent, exhibiting a uniform distribution across all outcomes with no detectable pattern or bias, making it comparable to the randomness of a physical dice. In terms of user experience, the device offered intuitive operation using a single button, clear LED display patterns, and satisfying tactile feedback. Furthermore, the reliability of the system was confirmed through extended testing, where it demonstrated stable operation without any failures.

SDG goals:

SDG 4 – Quality Education
SDG 9 – Industry, Innovation and Infrastructure
SDG 12 – Responsible Consumption and Production

Conclusion/ Summary

The **Digital Dice** project successfully demonstrates the practical use of **digital electronics** to replace a traditional dice with an electronic version. It uses a **555 timer**, **CD4017 counter**, and **74LS47 decoder** to generate and display random numbers from **1 to 6**. The circuit is **low-cost, reliable, and educational**, effectively showcasing concepts like multivibrators, counters, and decoders. Randomness was ensured through **high-frequency clocking**, and the design was verified through testing. Overall, the project provides a **simple yet powerful learning tool** and offers scope for future upgrades such as **FPGA implementation or microcontroller-based control**.

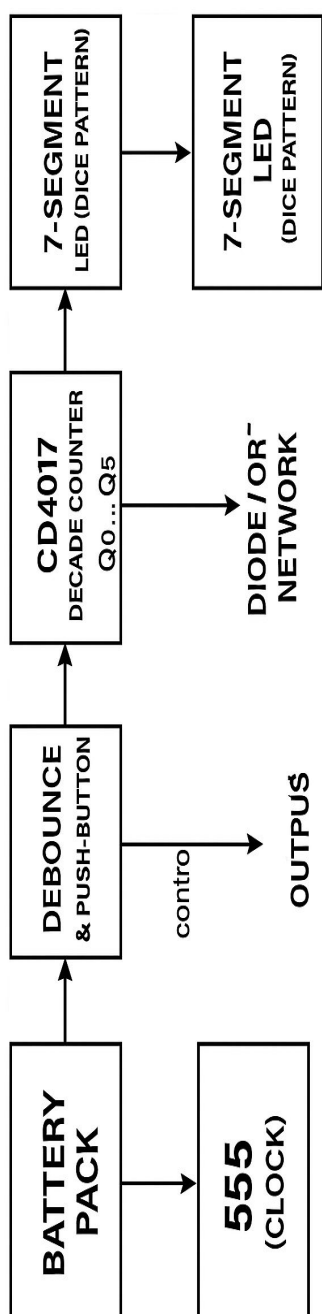
Contact Details : srajan.gupta2024@vitstudent.ac.in
devansh.pandey2024@vitstudent.ac.in
surya.singh2024@vitstudent.ac.in

dhanabal.r@vit.ac.in

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Methodology



The operation of the **Digital Dice** begins when the **push-button** is pressed, activating the **555 timer circuit**, which functions as a pulse generator. These clock pulses are sent to the **CD4017 decade counter**, which advances its output (Q0 to Q5) sequentially with each pulse. Each output of the counter represents a dice value from **1 to 6**. The counter outputs are then passed through a **diode OR network**, which combines the signals and encodes them into a suitable **binary-coded decimal (BCD)** form.

The BCD data is fed into the **74LS47 D-to-7-segment display driver**, which decodes the signals and lights up the corresponding segments on the **7-segment LED display** to show a number between 1 and 6, simulating the output of a dice. When the push-button is released, the clock pulses stop, freezing the counter at its current state. The number displayed on the LED represents the final random dice output. This process effectively mimics the behavior of a physical dice using digital electronic components.